LUCAS CRUZ

Senior Data Engineer | Gen. AI Automation Engineer

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Florianópolis, Santa Catarina, Brazil · Requires visa sponsorship

PROFESSIONAL SUMMARY

Senior Data Engineer and Gen. AI Automation Engineer with 5+ years building production data pipelines, AWS Medallion lakehouses, and LLM/n8n operational workflows across IoT, fintech, agribusiness, and automotive intelligence. Experienced in end-to-end delivery from Python/FastAPI/TypeScript services and governed datasets to containerized Azure/AWS deployments, CI/CD quality gates, and stakeholder-facing product delivery (English C1).

TECHNICAL SKILLS

Gen. AI & automation	LLM integration (Claude, Google AI Studio, xAI Grok), n8n, LangChain agents, prompt design, chatbots, Meta Cloud API, webhooks
Data engineering	Python, SQL, Apache Spark (PySpark), Pandas, Apache Airflow, Delta Lake, Parquet, Trino, Playwright, web scraping
Cloud, backend & delivery	AWS, Azure, GCP, OCI, Docker, Azure Container Apps, Bicep (IaC), FastAPI, TypeScript, NestJS, Next.js, Celery, Redis / Streams, Pydantic, Git, GitHub Actions (CI/CD), pytest
Databases & BI	PostgreSQL, SQL Server, Oracle Autonomous Database, Power BI, Qlik Sense, Apache Superset
Governance	LGPD/GDPR compliance, RBAC, data privacy, schema contracts (JSON Schema)
Labs (not client production)	dbt Core, Prefect, DuckDB, Snowflake
Languages	Portuguese (Native) · English (C1 – Advanced) · Spanish (B1 – Intermediate)

WORK EXPERIENCE

Gen. AI Engineer — TK Technologies | Curitiba, PR – Brazil (Hybrid) *Feb 2026 – Present · Contract*

- Built an omnichannel AI CRM (WhatsApp, Instagram, Facebook, Telegram, and email) with role-based access, Meta/Google Ads tracking, and LLM replies, deployed on Azure Container Apps with Bicep.
- Engineered a multi-source web scraping pipeline in Python targeting 10+ automotive parts websites, extracting pricing and product metadata at scale to deliver competitive market intelligence.
- Built n8n workflows for ETL, chat, messaging intake, and operational follow-up, using LLMs (Claude, Google AI Studio) and GCP APIs (Sheets, Drive).
- Developed a conversational AI chatbot that empowers business users to query and refine pricing datasets via natural language, reducing time-to-insight for non-technical stakeholders.
- Coordinated Notion-based delivery sprints, customer-facing demos, product training, sales support, and post-launch support; contribute to the TK web presence and ongoing CRM product development and maintenance.

Data Engineer — Spacecom Monitoramento | Curitiba, PR – Brazil (On-site) *Apr 2023 – Jul 2024*

- Designed and maintained an enterprise Big Data Lakehouse on AWS S3 leveraging Medallion Architecture (Bronze/Silver/Gold) alongside Delta Lake, Parquet, PySpark, and Trino to support heavy analytics workloads.
- Supported a high-availability IoT platform scaling to 95,000+ devices, efficiently processing hundreds of millions of telemetry records per day.
- Managed electronic security tag (RFID/NFC) data for government court systems, ensuring strict LGPD/GDPR compliance through robust data privacy controls, access governance, and secure handling of sensitive judicial data.
- Orchestrated and optimised ETL pipelines with Apache Airflow, reducing failure rates by ~30%; utilised Docker on AWS EC2 to cut deployment issues by an additional ~25%.

- Remediated inconsistencies in pipeline steps to boost downstream analytics accuracy by ~40%, successfully delivering reliable datasets to Qlik Sense and Apache Superset dashboards.

Data Engineer Consultant — wDiscover | Curitiba, PR – Brazil (Remote) *Jun 2022 – May 2024*

- Delivered end-to-end data projects across the fintech, agribusiness, and operations sectors by integrating high-volume payment providers (PagSeguro, Stone, Cielo), John Deere IoT APIs, and REST APIs.
- Designed dimensional Data Warehouses on SQL Server and Oracle Autonomous Database; automated complex ETL workflows with Airflow on Docker/Oracle Cloud to reduce manual effort by ~35%.
- Built dynamic Power BI dashboards to track critical KPIs, financial metrics, and operational performance for diverse client domains.

Data Analyst — Itaete Group | Curitiba, PR – Brazil (Hybrid) *Jun 2020 – Feb 2023*

- Designed a SQL Server Data Warehouse and robust Python/SQL pipelines using Pandas to cleanse, transform, and enrich data from IoT devices, PostgreSQL, APIs, and CSV sources.
- Built Power BI dashboards and automated legacy Excel/VBA reporting workflows, slashing data processing time by ~50% and significantly reducing manual operational workloads.
- Developed internal systems for inventory management and financial automation, streamlining invoice generation and automated weekly report distribution.

NOTABLE PROJECTS

Omnichannel AI CRM · *NestJS, Next.js, PostgreSQL, LLMs, Meta & Google Ads APIs, Azure Container Apps, Docker · TK Technologies – Jun 2026 – Present*

- Built a secure omnichannel CRM integrating WhatsApp, Instagram, Facebook, Telegram, and Email into a unified inbox with role-based access control (RBAC).
- Layered Meta/Google Ads performance data and LLMs (Claude/Google AI Studio) for auto-replies, outbound messaging assistance, and automated campaign analysis.

Car Parts Price Searcher · *Python, FastAPI, Celery, Playwright, Redis, PostgreSQL, n8n, Azure, Bicep · TK Technologies – Feb 2026 – Present*

- Built a car-parts price searcher across 10+ sites, with parallel public-price and internal-stock lookup, auditable reporting, and notifications from containerised Azure services.

IoT Data Lakehouse on AWS – Medallion Architecture · *AWS S3, Apache Spark, Delta Lake, Airflow, Trino, Apache Superset · Client Project – Feb 2024*

- Designed a Bronze/Silver/Gold lakehouse on AWS S3 ingesting from APIs and databases. Virtualised the Gold layer with Trino and surfaced governed datasets through BI dashboards.

EDUCATION

Bachelor of Systems Analysis and Development — Dom Bosco University, Curitiba, Brazil · Jun 2023

Majors: Information Systems · Business Analytics · Data Analysis · Database Systems · Software Engineering

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LICENSES & CERTIFICATIONS

- IELTS General — English C1
- AI for Writing and Communicating — Google (Credential ID: HE8YE5IS7WVW) · Jul 2026
- AI for Research and Insights — Google (Credential ID: 4PZDF3XQOF55) · Jul 2026
- AI for Brainstorming and Planning — Google (Credential ID: 7SFW2HGOKIU5) · Jul 2026
- AI Fundamentals — Google (Credential ID: UKBGM0AZUIS2) · Jul 2026